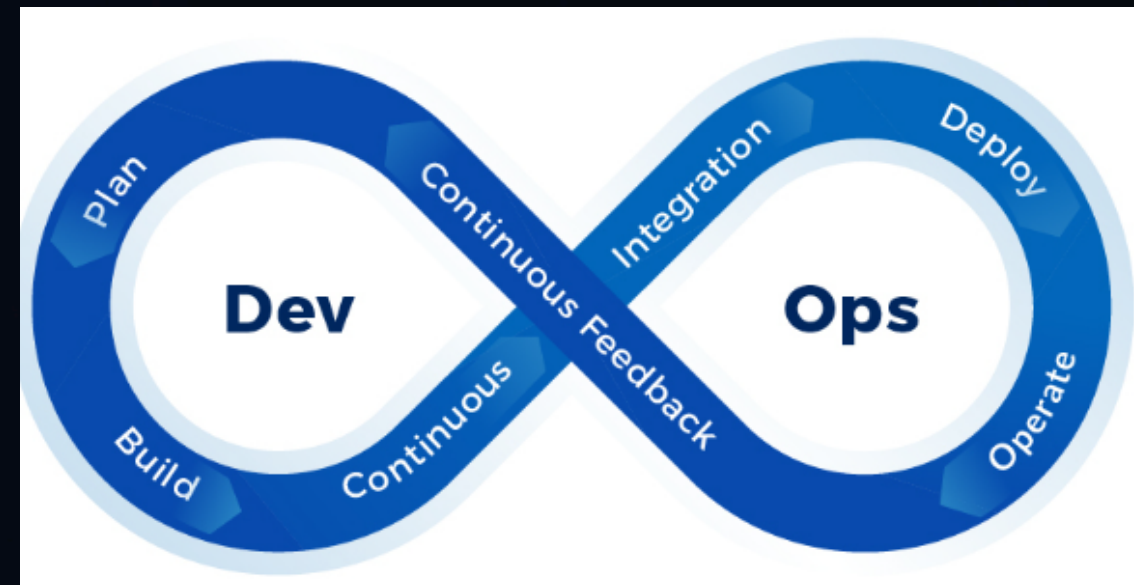


Deploy Detective:

AI-Powered Incident
Response for Software &
DevOps Teams

Investigate, Correlate, Rollback, Verify
— Transparency

Windwalkers



The Production Incident Problem

Sudden production slowdowns or failures impact user experience and revenue

Traditional dashboards show symptoms but lack root-cause clarity

Manual investigation delays resolution and risks unsafe fixes

ERROR RATE

6.42%

Last 15 minutes



P95 LATENCY

1.82s

Last 15 minutes



ACTIVE CONNECTIONS

12,358

Last 15 minutes



What Deploy Detective Does

An AI-powered agent that investigates, responds, and verifies production incidents end to end.



Transparent Agent Investigation Loop

The agent autonomously queries logs, metrics, and deployment events in Exasol

Each step produces auditable SQL queries and evidence cards

Decisions are based on deterministic, verifiable data

No black boxes. Every action is traceable.

 **Deploy Detective** Autonomous SRE

● INC-2026-0042

● Exasol V8

 Reset

 Guided Replay

 1. Anomaly

 **2. Agent SQL**

 3. Root Cause

 4. Remediation

 5. Proof Ledger

Autonomous Reasoning & Exasol SQL Trace

Deterministic incident analysis verified through in-memory SQL execution

4 Queries Executed · 86ms Avg

01 UNDERSTAND INCIDENT **Completed**

0.1s

User reports checkout is slow. Mapped customer-reported errors to service checkout in prod-us-east-1 within the 14:15-14:35 UTC evaluation window.

02 QUERY EXASOL: ERROR-RATE TIMELINE **Completed**

87ms · 2.0M rows

Aggregated 2.0M log rows with millisecond-level windowing. Identified exact inflection at 14:32:06 UTC.

```
SELECT service_id, DATE_TRUNC('MINUTE', ts) AS minute, SUM(CASE WHEN level = 'ERROR' THEN 1 ELSE 0 END) / COUNT(*) AS error_rate FROM logs WHERE service_id = 'checkout' GROUP BY service_id, DATE_TRUNC('MINUTE', ts) ORDER BY minute;
```

Engine: Exasol Personal · In-Memory

Execution time: 87ms

03 CORRELATE DEPLOYMENTS **Completed**

94ms [View SQL](#)

Found deployment **DEP-1842 (checkout v2.4.0)** — exactly 6 minutes before error spike.

04 CHECK DEPENDENCIES **Completed**

71ms

Downstream dependency payment p95 latency increased from 112ms to 628ms following checkout deploy.

Live Investigation Trace

with Exasol

SQL Evidence

Agent analyzes error-rate timeline, correlates deployments and connection pool metrics

Live SQL queries executed in-memory for instant insight

Every finding linked to deployments, latency, and connection activity

✓ 1. Anomaly

✓ 2. Agent SQL

✓ 3. Root Cause

✓ 4. Remediation

5 5. Proof Ledger

Deterministic Proof Ledger & Post-Remediation Verification

Real-time telemetry validation verified over 120s rolling window via Exasol in-memory analytics engine.

● SL0 Compliance Restored: 99.99%

TELEMETRY COMPARISON & RECOVERY VERIFICATION MATRIX

Error Rate

✓ Verified Healthy

34.2% → 0.12%

Evaluation: 480k Envoy ingress traces (0 anomalies detected)

P95 Latency

✓ Target < 150ms Met

1,840ms → 118ms

Evaluation: Tail latency normalized across all 12 worker pods

Active DB Connection Pool

✓ Healthy Headroom

100% Saturation → 24%

Evaluation: Connection pool exhaustion cleared in v2.3.9

Ingress 503 HTTP Count

✓ Cleared

1,420 / min → 0 / min

Evaluation: Zero 5xx responses recorded over trailing 120s window

EXASOL IN-MEMORY AUDIT & CRYPTOGRAPHIC PROOF LEDGER

● Immutable Ledger Stored

Verification Query Executed:

SELECT service_id, COUNT(*) AS samples, SUM(CASE WHEN status ≥ 500 THEN 1 ELSE 0 END) AS errors FROM prod_traces WHERE service_...

Exasol Personal In-Memory · 42ms on 480k rows

Audit Hash (SHA-256):

e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855

📄

Action Signature:

agent://deploy-detective/v2.4-controller (Ed25519 verified)

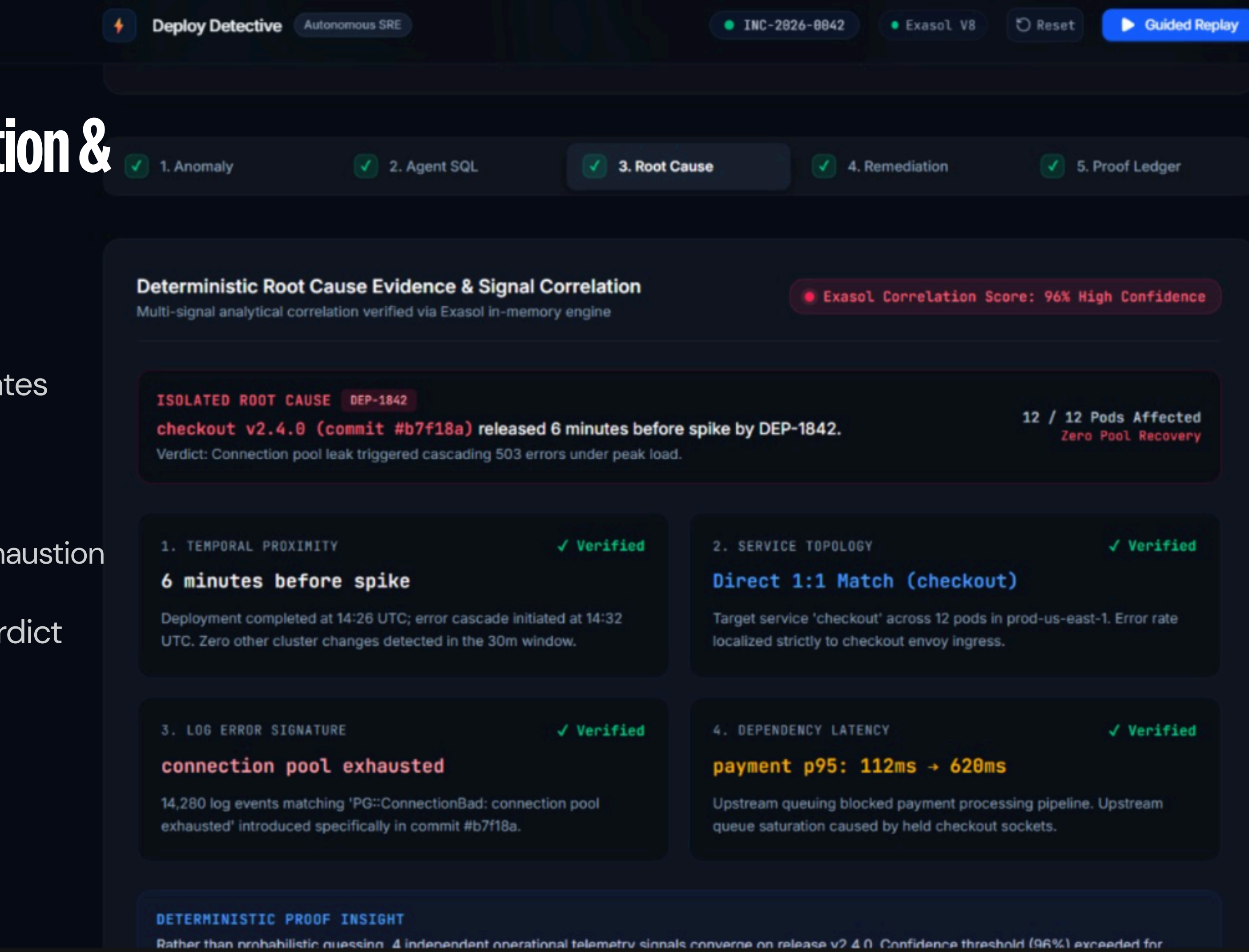
Root-Cause Correlation & Evidence

Multi-signal correlation isolates the faulty deployment

Error spike aligns with recent deployment and resource exhaustion

Deterministic root cause verdict with confidence scoring

Actionable evidence for faster, safer resolution

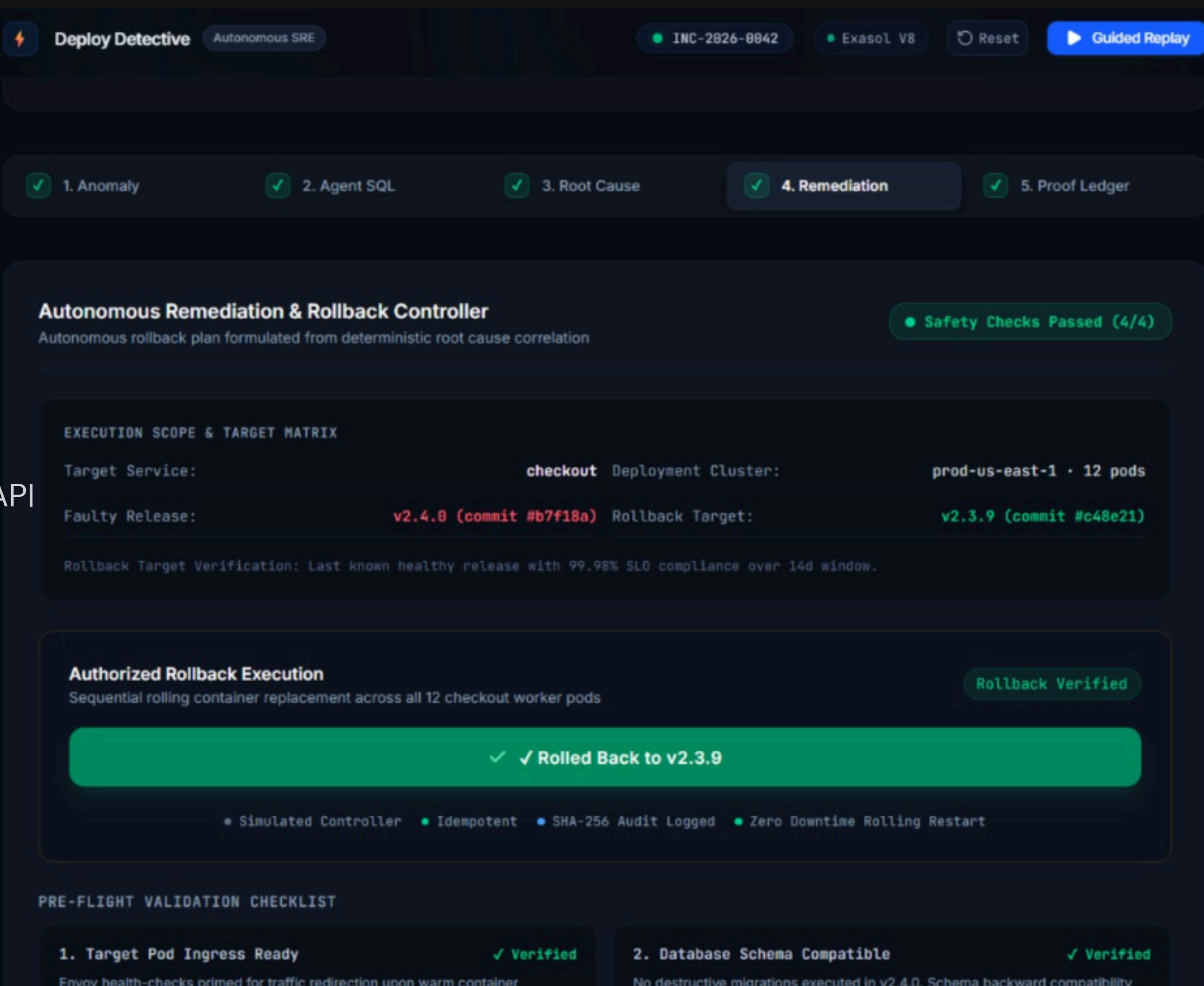


Safe Rollback & Verification

Agent triggers rollback to last stable release via deployment API

Safety checks ensure schema compatibility, ingress readiness, and audit logging

Post-rollback metrics confirm recovery and SLA compliance



Why Exasol is Essential

Real-time SQL investigation across logs, metrics, and deployment events

Multi-signal correlation for deterministic root cause analysis

Efficient querying powers autonomous reasoning and audit transparency

In-memory performance at scale for fast, reliable decisions

TOP ENDPOINTS BY LATENCY

ENDPOINT	P95 LATENCY
/checkout	4.21s
/api/report	3.18s
/search	2.74s
/api/user	2.31s
/dashboard	1.98s

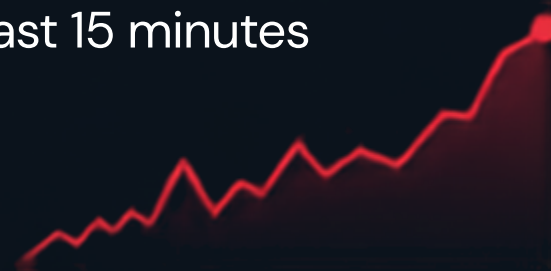
ACTIVE CONNECTIONS

1,892
↑ 1,102 vs 1h ago

ERROR RATE

6.42%

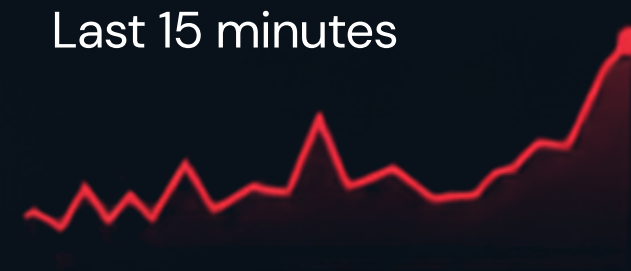
Last 15 minutes



P95 LATENCY

1.82s

Last 15 minutes



ACTIVE CONNECTIONS

12,358

Last 15 minutes



Ingest

logs, metrics,
events

Query

real-time
SQL at scale

Correlate

across signals
instantly

Resolve

root cause with
confidence

Audit

every step,
end to end

Impact, Differentiation & Responsible Guardrails

Maximum Impact

Autonomous, data-driven incident response reduces downtime

Total Transparency

SQL and evidence-backed investigations build trust, not black-box AI

Safe Automation

Bounded rollback actions ensure safe, auditable remediation

Verified Outcomes

Proof ledger validates post-remediation health and SLA adherence

Deploy Detective

Closing Demo Takeaway

Delivers fast, transparent, and safe incident resolution

Leverages Exasol's analytics to turn telemetry into actionable insights

See the SQL, evidence, and remediation logic — not a black box

Ready to transform incident response with trusted AI-driven autonomy

Thank you — happy to demo and discuss next steps.

